SQL DATA SCIENCE CAPSTONE PROJECT

SportStats: 120 Years of Olympic History



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Preparation & Setup

- **Client Selection:** Why SportsStats
- **Data Engineering:** Ingesting 271k rows, cleaning nulls, and building SQLite database
- **Initial Exploration:** Dataset scale, 120-year span, and female age trends
- **Relational Model:** Designing the Star Schema ERD

Client & Dataset Selection

- **Client Chosen: Client 3 — SportsStats**
- **Dataset:** 120 Years of Olympic History (`athlete_events.csv`, 271,116 records).
- **Why: SportsStats** delivers insights to two main audiences: **local news media** (looking for historical narratives and regional achievements) and **elite personal trainers** (seeking health, aging, and biometric benchmarks). This dataset spans 120 years (1896–2016) and contains physical metrics, event outcomes, and national metadata, making it ideal for answering both media and athletic performance queries.

STEP 1: PREPARATION & SETUP

Data Import & Cleaning Steps

1. Extract / 📥 CSV Ingestion

- Imported **271,116** raw records into a **Pandas DataFrame.**
  - Validated schema, initial row counts, and data structure from **`athlete_events.csv`**

2. Clean / 🛠️ Nulls & Types

- Imputed missing values & standardized data types.
  - Replaced **85.3%** **`NaN`** medals with **`'No Medal'`**; cast **`Year`** to **`int`** and **`Age`** to nullable **`Int64`**.

3. Transform / 📐 Feature Engineering

- Engineered a custom **BMI** metric for trainer profiling.
  - Enabled biometric benchmark analysis across different sports.

4. Load / 💾 SQLite Export

- Persisted the clean dataset into **`olympics.db`**.
  - Created relational tables via Python's native `sqlite3` driver for downstream SQL queries.

STEP 1: PREPARATION & SETUP

Initial Data Exploration & Key Statistics

total_records	unique_athletes	total_countries	total_sports	missing_age_count	missing_height_count	missing_weight_count	total_medals_awarded
271116	135571	230	66	9474	60171	62875	39783

We can see we have pretty thorough data across 271,116 records with complete demographic coverage.

Era	Season	Sex	Medalists	Avg_Age	Min_Age	Max_Age
1896-1919	Summer	F	52	29.7	15	63
1896-1919	Summer	M	2163	27.3	10	71
1920-1959	Summer	F	495	22.8	11	69
1920-1959	Summer	M	4761	27.8	12	73
1920-1959	Winter	F	63	23.3	15	39
1920-1959	Winter	M	619	27.2	15	58
1960-1999	Summer	F	2882	23.2	13	52
1960-1999	Summer	M	7151	25.4	13	66
1960-1999	Winter	F	459	24.4	13	42
1960-1999	Winter	M	1323	25.5	14	50
2000-Present	Summer	F	3251	25.6	15	52
2000-Present	Summer	M	4055	26.6	15	61
2000-Present	Winter	F	594	26.0	15	46
2000-Present	Winter	M	791	27.6	15	49

We can see as more women were allowed to compete, the gap between male and female medalists narrowed significantly over time.

STEP 1: PREPARATION & SETUP

Initial Data Exploration & Key Statistics

Sport	Sex	Avg_Age	Avg_Height_cm	Avg_Weight_kg	Avg_BMI
Athletics	F	25.1	169.3	60.2	20.9
Athletics	M	25.4	179.8	73.8	22.7
Equestrianism	F	34.2	167.9	58.6	20.8
Equestrianism	M	34.1	176.6	70.9	22.7
Gymnastics	F	18.7	156.2	47.8	19.5
Gymnastics	M	23.8	167.6	63.3	22.5
Swimming	F	19.6	171.5	61.5	20.9
Swimming	M	21.5	184.4	78.0	22.9
Weightlifting	F	24.0	160.5	67.8	26.0
Weightlifting	M	25.5	169.2	81.7	28.0

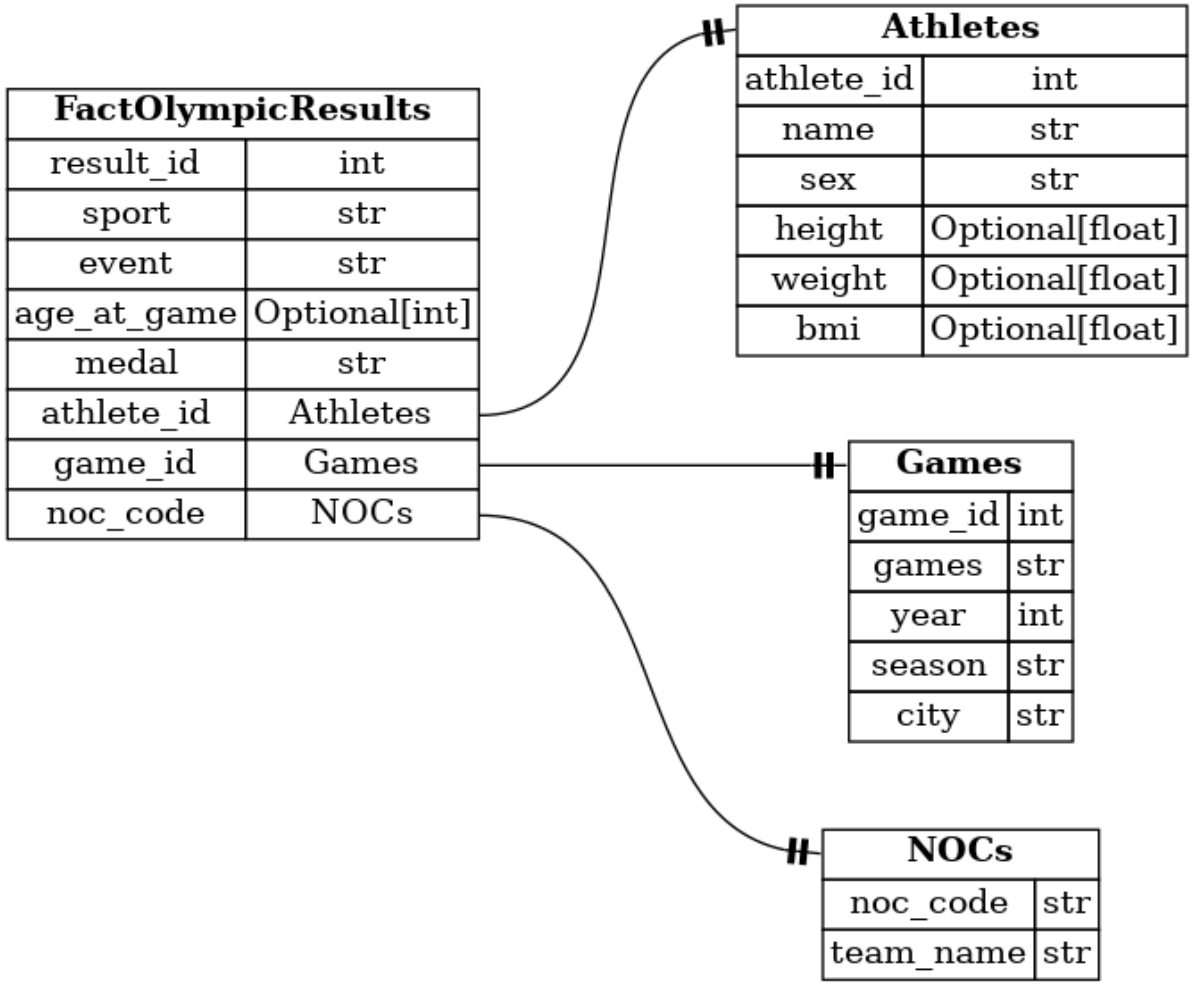
Biometric Clustering

- **Weightlifting leads BMI** (F: 26.0, M: 28.0), reflecting high muscle density requirements.
- **Gymnastics recorded the lowest BMI** (F: 19.5, M: 22.5), prioritizing agility and a high power-to-weight ratio.
- **Equestrianism showcases peak athletic longevity**, averaging ~34.1–34.2 years of age.

Target Audience Application

- **Essential for Personal Trainers** establishing baseline biometric target profiles across distinct sport disciplines.
- **Demonstrates practical business utility** of engineered features (BMI) within our ETL pipeline.
- **Supports multi-persona analysis**, serving both sports media coverage and athletic fitness modeling.

Proposed Entity Relationship Diagram (ERD)



Part I: Project Proposal

Description:

- Scope & dual-audience context (Media vs. Trainers).

Questions:

- 3 key research questions driving data analysis.

Hypotheses:

- Testable assumptions on biometrics, age, and country dominance.

Approach:

- SQL/Pandas analytical pipeline and metrics.

Description

- **Scope & Dataset:** Analyzes 120 years of Olympic history (1896–2016) to uncover long-term trends in athletic performance, country dominance, and athlete biometrics.
- **Core Objective:** Transforms raw historical records into actionable insights across demographic, geographic, and sport-specific dimensions.
- **Dual Target Audiences:**
  - **Sports Media Analysts:** Seeking high-impact historical narratives and comparative country dominance.
  - **Personal Trainers & Athletic Coaches:** Looking for sport-specific physical benchmarks and profiling.
- **Feature Engineering Concept:** Plans to leverage engineered metrics like Body Mass Index (**BMI**) to bridge macro-level historical trends with micro-level biometric profiling.
- **Intended Deliverables:** A structured SQLite database paired with a data visualization pipeline to enable targeted, data-driven querying for both audiences.

Questions

- **Q1 (Media & History):** How has country dominance evolved over time across Summer and Winter Games, and which nations consistently lead gold medal counts in major sports?
- **Q2 (Biometrics & Performance):** How do physical attributes (Height, Weight, BMI) and average peak ages differ across athletic disciplines, and how have these biometric baselines shifted over time?
- **Q3 (Demographics & Participation):** How has female participation and medal representation grown relative to male participation across different Olympic eras and sport categories?

Hypotheses

- **H1 (Biometric Specialization):** Athletes competing in power and strength sports (e.g., Weightlifting, Judo) maintain significantly higher average **BMI** levels than athletes in agility- and endurance-based sports (e.g., Gymnastics, Distance Running).
- **H2 (Age & Athletic Peak):** High-impact explosive sports (e.g., Gymnastics, Swimming) favor younger athletes with earlier peak performance ages, whereas endurance or skill-dominated sports (e.g., Equestrianism, Shooting) exhibit higher average ages and longer career spans.
- **H3 (Geographic Dominance Shifts):** Top-tier medal distribution has become increasingly diversified over the modern era (2000–Present) compared to early 20th-century Games, which were heavily dominated by a small group of Western nations.

Approach

- **Analytical Pipeline:** Query the local SQLite database (`olympics.db`) using Python and Pandas to evaluate relationships across key feature sets.
  - **Phase 1 — Biometric Baselines:** Analyze demographic and physical attributes (`Age`, `Height`, `Weight`, `BMI`, `Sex`) grouped by `Sport` and `Event` to establish baseline profiles for trainers and performance coaches.
  - **Phase 2 — Temporal & Geographic Tracking:** Perform aggregations across `Games` (`Year`, `Season`) and `NOCs` (`team_name`, `noc_code`) to track historical participation growth and regional medal distributions.
- **Evaluation Metrics:** Utilize summary statistics (mean, median, IQR), historical growth rates, and proportional rankings (top-N country distributions).
- **Exploratory Visualizations:** Map data relationships using distribution plots, multi-line trend charts across eras, and sport-by-sport correlation matrices.
- **SQL View Optimization:** Build SQL aggregation views to streamline dynamic querying for both sports media stories and fitness profiling.

SQL DATA SCIENCE CAPSTONE FINAL PRESENTATION

Part II: Execution, Architecture & Analytical Insights



Fom Proposal Hypotheses to Data Modeling & Metric Analysis

Part II Agenda & Technical Focus

Data Normalization & Architecture

- Converted flat dataset into a 5-Table Star Schema.
- Deduplicated team-sport medal counts and unified historical country code fragmentation (e.g., URS -> RUS).

Feature Engineering & New Metrics

- **Event-Level Medal Efficiency Ratio /eMER/:** Measures real delegation efficiency without relay/team medal inflation.
- **ΔBMI Score:** Measures individual athlete physical deviation from their sport's baseline norm.

Hypothesis Testing & Parametric Fits

- Evaluated core biometric hypotheses and fitted OLS linear models using Pandas and SQL queries.

Additional Analytical Questions

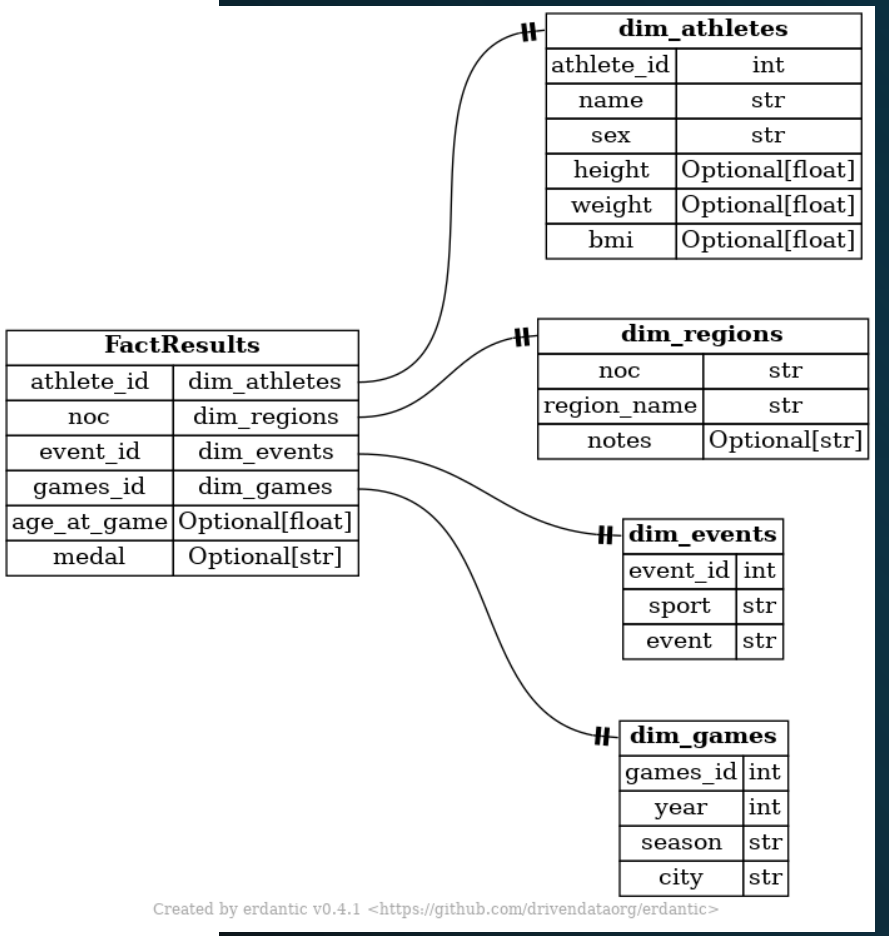
- **Delegation Efficiency:** Hyper-specialized small nations lead event-level MER (Cyprus 1.581, Iceland 1.580, Georgia 1.529).
- **Biometric Evolution:** Tracked historical widening of BMI gap across eras and identified top ΔBMI outliers in Judo and Weightlifting.

Strategic Recommendations & Conclusion

- Actionable applications for Sports Media Analysts (eMER reporting) and Athletic Coaches (ΔBMI diagnostic baselines).

Data Architecture & Normalization

- **Initial Baseline:** Proposed 4-table ERD during Milestone 1 exploration.
- **Data Engineering Challenges:** Solved team-sport medal inflation (e.g., relay golds) and historical NOC fragmentation (e.g., URS vs. RUS).
- **Final 5-Table Star Schema:** Introduced dim_events for event-level deduplication and dim_regions for unified regional mapping.



Data Overview

Metric	Count	Mean	Std	Min	50% (Median)	Max	Skew
Age	261642.0	25.56	6.39	10.0	24.0	97.0	N/A
Height	210945.0	175.34	10.52	127.0	175.0	226.0	0.02
Weight	208241.0	70.70	14.35	25.0	70.0	214.0	0.80
BMI	206853.0	22.78	2.91	8.36	22.53	63.90	1.40

- **Distribution & Skewness: Height** exhibits a near-perfect normal distribution (Mean = 175.3 cm, Skew = 0.02). In contrast, **Weight** (Skew = 0.80) and **BMI** (Skew = 1.40) show heavy right-skewness due to strength disciplines.
- **Extreme Ranges: Age** spans 10 to 97 years (Mean = 25.6), reflecting youth gymnasts vs. equestrian veterans. **BMI** spans 8.36 to 63.90 (Mean = 22.8).
- **Data Quality & Purpose:**
  - Validated raw distributions and flagged non-athletic noise (e.g., historical *Art Competitions*).
  - Established physical baselines prior to Star Schema feature engineering.

Feature Engineering & New Metrics

- Medal Efficiency Ratio (**MER**) vs. Event-Level Medal Efficiency Ratio (**eMER**)
  - Athlete-Level Medal Efficiency Ratio (MER):
    - Formula: **MER = Total Medals Won / Unique Athletes Sent**
    - Purpose: Measures raw national yield per athlete, but remains susceptible to team-sport inflation.
  - Event-Level Medal Efficiency Ratio (**eMER**):
    - Formula**: eMER = Deduplicated Event Medals Won / Unique Athletes Sent**
    - Purpose: Normalizes country performance by counting 1 team medal per event (e.g., 1 relay gold = 1 event medal). This isolates true national efficiency and reveals hyper-specialized, high-yield delegations (e.g., Ethiopia in distance running, or Georgia in combat sports) compared to massive, high-volume delegations.
- BMI Deviation Score (**ΔBMI**)
  - Formula:**Δ BMI = Athlete BMI - Sport Average Baseline BMI**
  - Purpose: Quantifies how far an individual athlete's body composition departs from their specific discipline's baseline norm. Serves as a key diagnostic metric for athletic coaches and personal trainers to evaluate physical outliers and specialized positional roles.

Sport	Sex	Average BMI
Gymnastics	F	19.52
Gymnastics	M	22.53
Swimming	F	20.87
Swimming	M	22.92
Athletics	F	20.91
Athletics	M	22.69
Weightlifting	F	25.96
Weightlifting	M	28.04

Hypothesis 1: Biometric Specialization

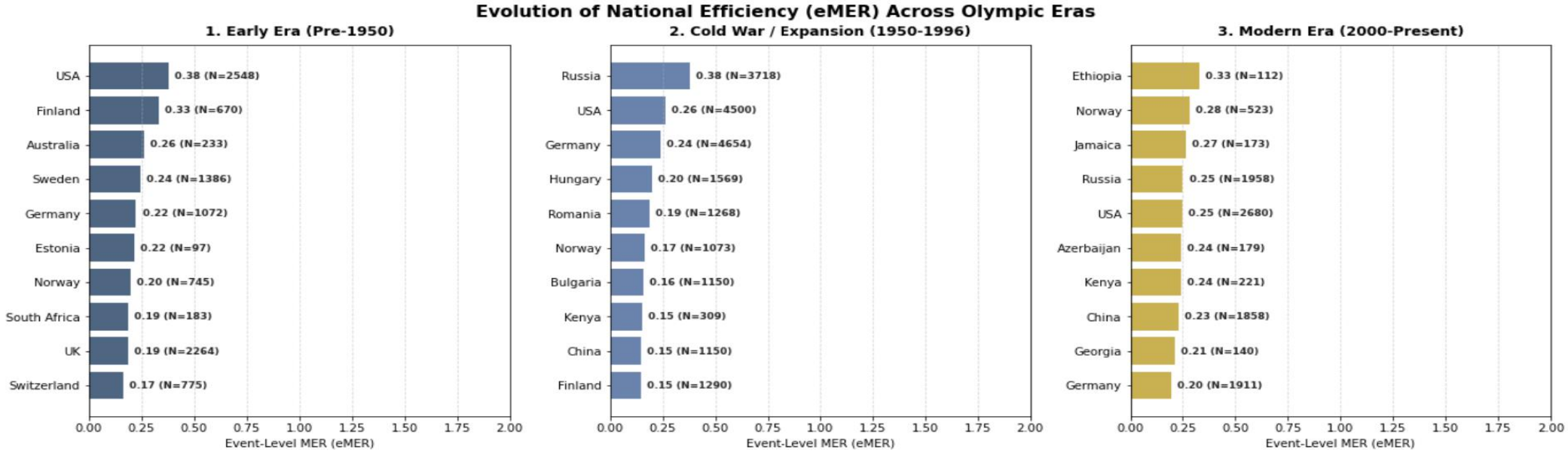
SQL aggregations confirm clustering by sport discipline. Mainstream speed and endurance disciplines (**Athletics** & **Swimming**) establish a consistent baseline of **20.87 - 22.92 BMI**. In contrast, power-based **Weightlifting** (**28.04** M / **25.96** F) averages over **5.5 to 6.4 BMI points higher** than agility-focused **Gymnastics** (**22.53** M / **19.52** F), fully validating Hypothesis 1.

Sport	Medalis	Avg_Age	Youngest	Oldest
Swimming	1637	20.9	12	46
Gymnastics	1252	23.4	10	45
Weightlifting	500	25.3	17	40
Shooting	734	33.4	16	72

Hypothesis 2: Age & Athletic Peak

Proven. SQL aggregations demonstrate a **12.5-year gap** in average medalist age between early-peak disciplines like **Swimming** (20.9 yrs; youngest 12) and technical precision disciplines like **Shooting** (33.4 yrs; oldest 72). High-impact agility and aquatic sports favor younger athletes, whereas technical skill sports offer significantly extended athletic longevity.

Hypothesis 3: Geographic Dominance Shifts

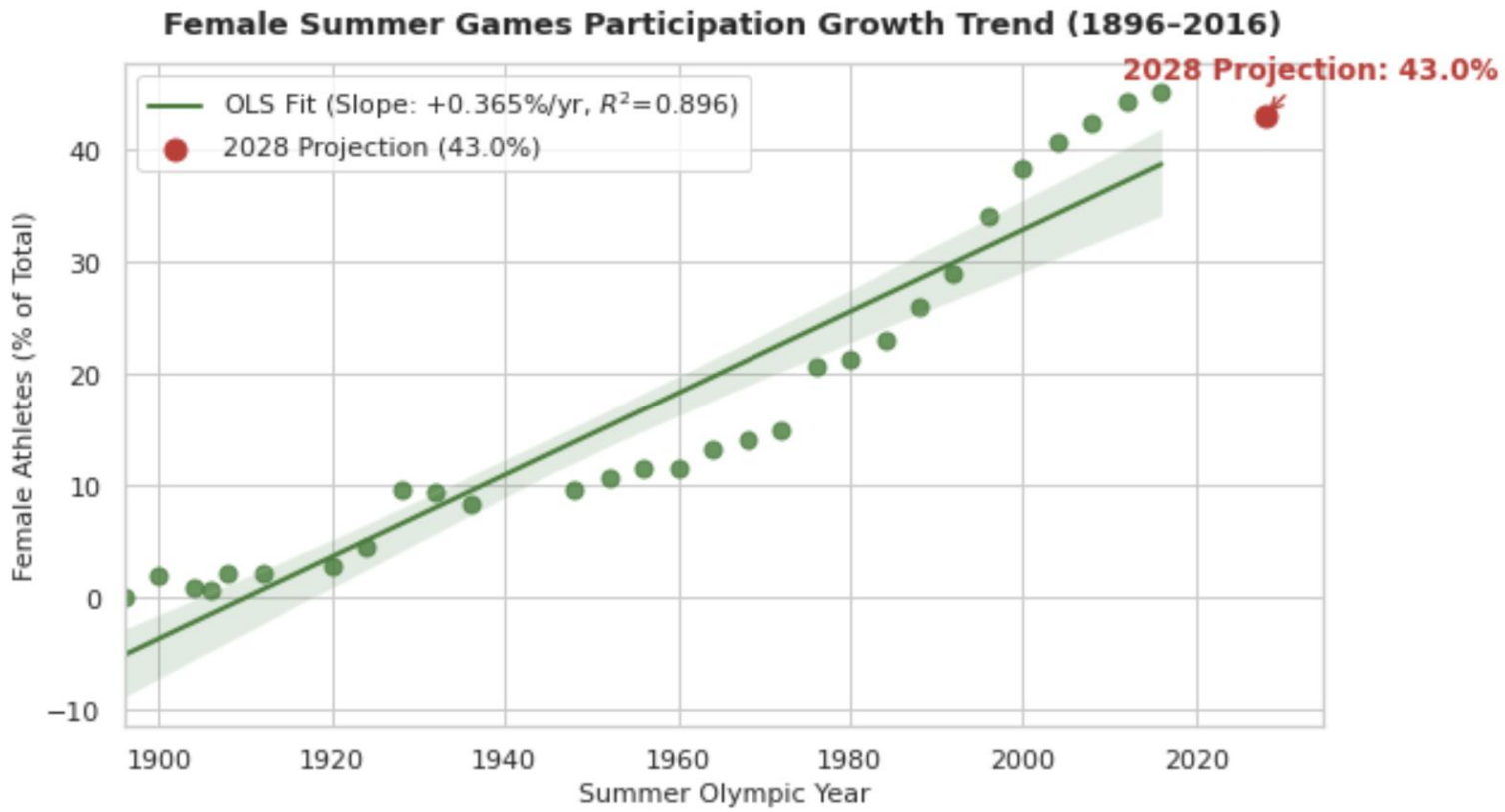


•**Proven.** Top-tier Olympic efficiency evolved from Early Western dominance through Cold War era volume dominance—where only huge delegations achieved high efficiency—to modern global hyper-specialization. Today, smaller, focused delegations like Ethiopia, Jamaica, and Georgia lead the world in yield per athlete sent.

Additional Analytical Insights

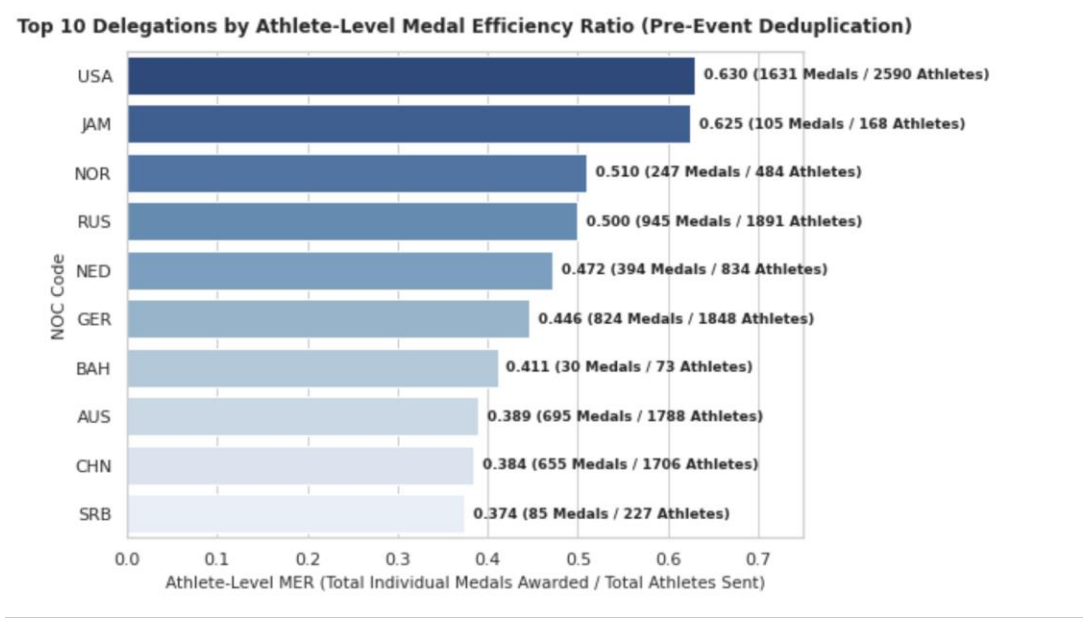
- **Macro Trends & Demographics:** Tracking female participation trajectories (OLS fit) and physical attribute correlations.
- **Metric Normalization (eMER vs MER):** Contrast between athlete-level volume inflation and event-level national yield.
- **Biometric Outlier Profiling:** Historical BMI divergence across sports and individual physical outliers (**ΔBMI**).

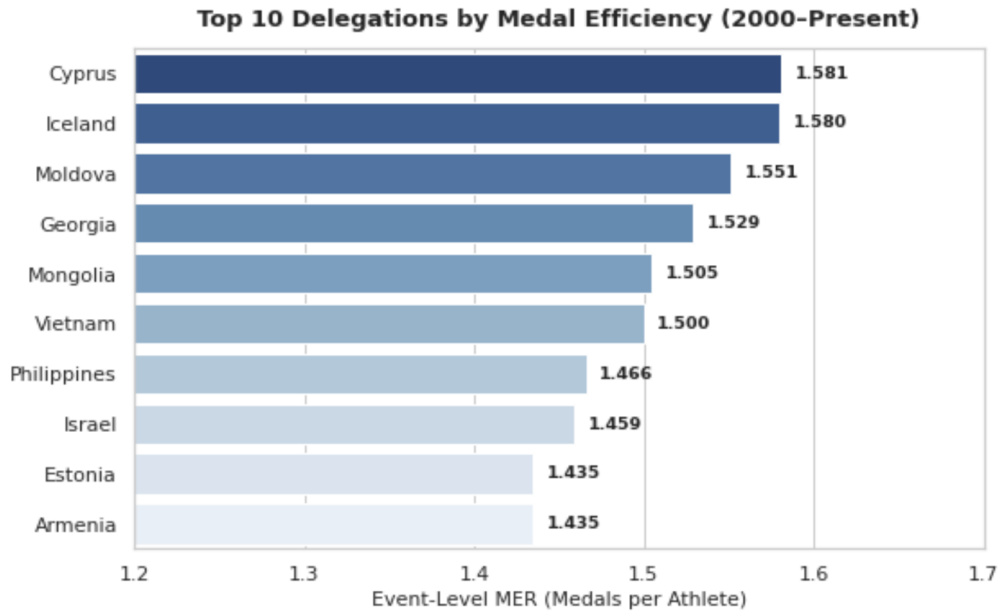
Demographic Expansion: Female Participation Trajectory



Fitting an **OLS** regression on historical Summer Games data confirmed a strong, steady linear progression toward gender parity over time, projecting a historical linear baseline of 43.0 % female athlete participation for the 2028 Games. Though recent IOC targets (50%) highlights how proactive policy in the modern era accelerated growth faster than historical linear rates alone.

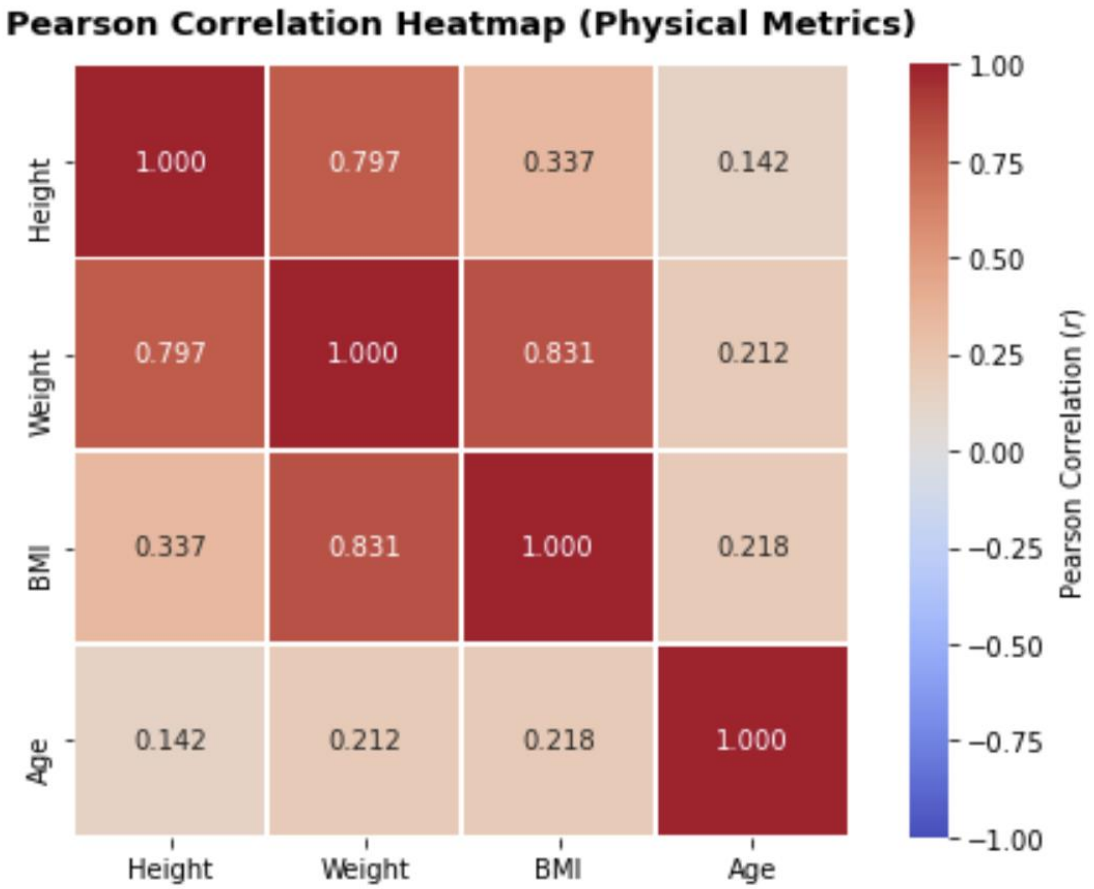
Metric Normalization: Athlete-Level MER vs. Event-Level eMER





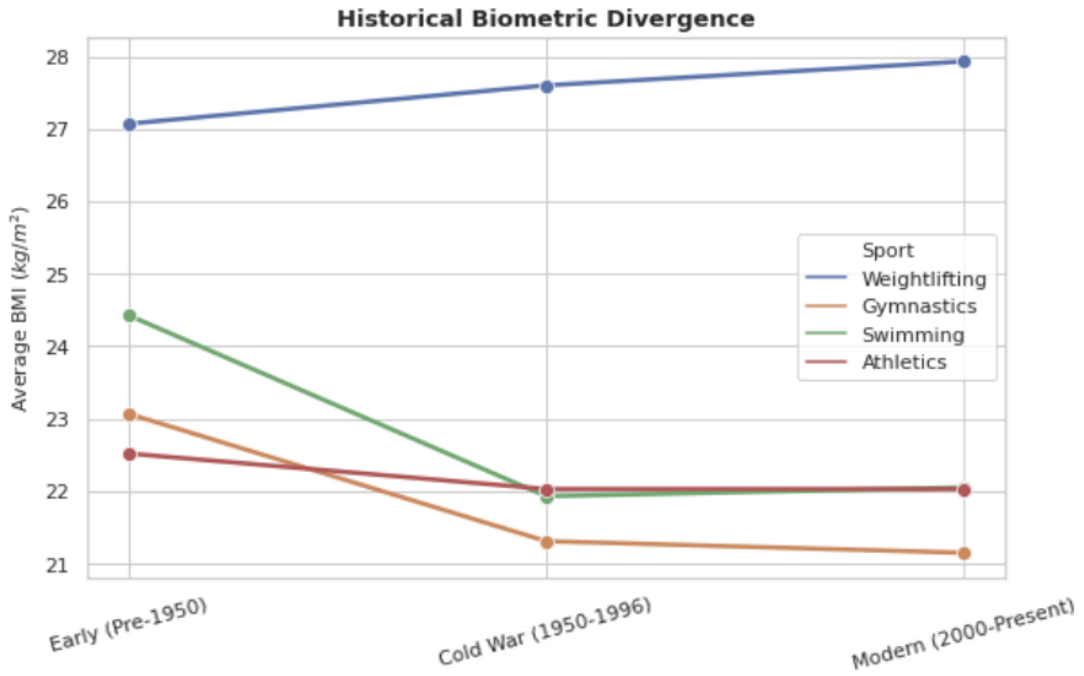
Raw MER is inflated by large team-sport rosters (USA, Russia, Germany). Deduplicated eMER isolates hyper-specialized, high-yield delegations like **Cyprus (1.581)**, **Iceland (1.580)**, and **Moldova (1.551)**.

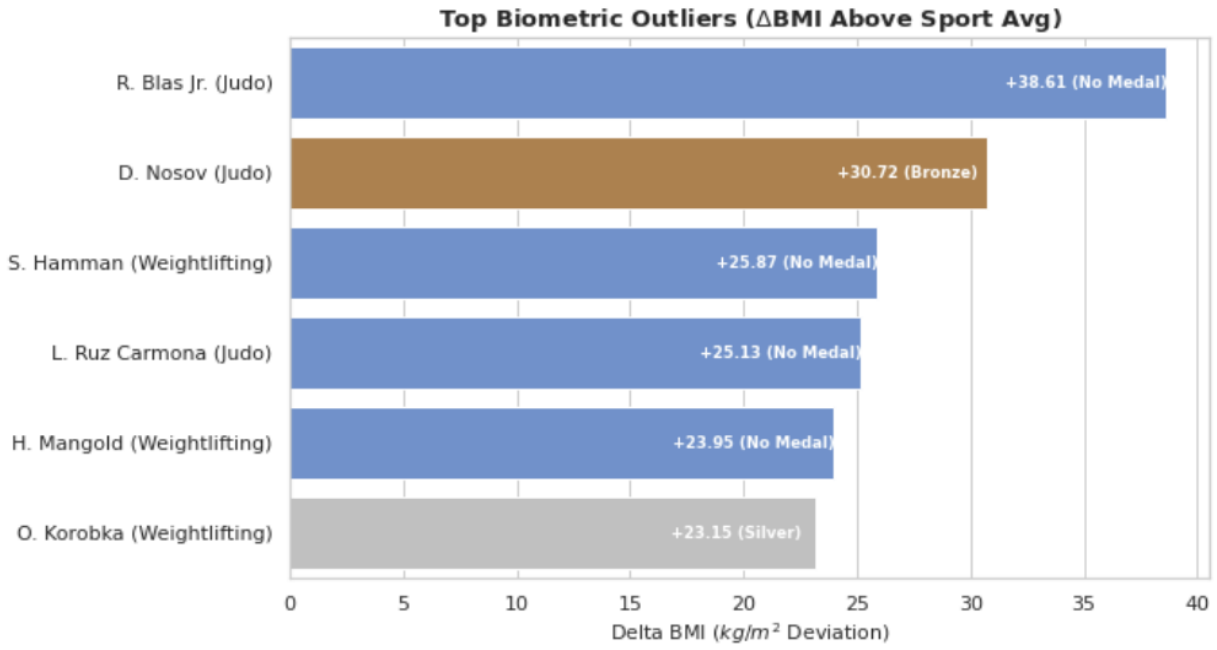
Feature Dependencies: Pearson Correlation Matrix



Weight vs. BMI (r = 0.831) overwhelms Height vs. BMI (r = 0.337), proving high BMI is driven by muscle density rather than stature. Age independence (r ~ 0.14 - 0.22) confirms body types are sport-selected, not age-dependent.

Biometric Specialization & Physical Outlier Profiling





Historical biometric analysis reveals that Weightlifting steadily expanded its body mass baseline over time, while Gymnastics and Swimming experienced a parallel drop toward leaner modern baselines and Athletics remained remarkably stable. Heavyweight combat and strength disciplines exhibit extreme individual positive deviations up to +35 kg/m^2 above sport norms, demonstrating how modern training has driven extreme morphological specialization.

Conclusion

- **Hypothesis 1 (Biometric Specialization - Proven):**
  - Physical profiles are strictly discipline-bound. Power-based Weightlifting maintains a 5.5 to 6.4 BMI point elevation over speed and agility disciplines like Gymnastics and Swimming.
- **Hypothesis 2 (Age & Athletic Peak - Proven):**
  - SQL analysis reveals a 12.5-year gap in average medalist age between early-peak aquatic sports and technical precision disciplines.
- **Hypothesis 3 (Geographic Dominance Shifts - Proven):**
  - Top-tier Olympic efficiency shifted from Cold War era volume dominance to modern global hyper-specialization. Today, smaller, focused delegations like Ethiopia, Jamaica, and Georgia lead the world in yield per athlete sent.
- **Feature Engineering & Metric Normalization:**
  - Raw athlete-level MER is distorted by team-sport size. Deduplicated event-level MER (eMER) correctly isolates true national efficiency, while Delta BMI serves as an effective diagnostic tool for evaluating physical outliers.

Recommendations & Next Steps

- **For Athletic Directors & Personal Trainers / Coaches:**
  - Target Outlier Diagnostics (**ΔBMI**): Utilize sport-specific **ΔBMI** baselines rather than generic BMI scale targets to compare client physical builds and optimize muscle mass and weight-class suitability.
  - Focus Specialization Strategy: Allocate delegation resources into high-yield, sport-specific niches high eMER athlete efficiency.
  - Track Era Benchmarks: Recognize that physical requirements evolve.
- **For Sports Media Analysts & Journalists:**
  - Adopt Event-Level MER (eMER): Shift from raw medal counts to deduplicated event-level efficiency to accurately report on high-performing smaller nations and eliminate team-sport inflation.
  - Contextualize Biometric Evolution & Eras: Feature narrative storylines around physical evolution and reference age-peak gaps when evaluating athletic longevity.

Questions & Discussion

Thank you for your time. Open for questions and feedback

